

A, B, C). Analysis

The two AIs I used for this assignment is Chatgpt and Claude. I accessed Chatgpt through my computer and Claude through my computer. The prompt I used was: Write a C program that picks a secret number between 1-10. The user only gets 3 guesses. Other rules: must use a fixed secret number, use Printf to prompt and scanf to read the guess, use a While or For loop to let the user guess 3 times, Use if-else statements to check if the guess is too high or low or correct, stop the loop immediately once the guess is correct, and print a final win or lose message.

D).

Chatgpts Code:

```
#include <stdio.h>
```

```
int main() {
    int secretNumber = 7;
    int guess;
    int attempts = 0;

    printf("Guess the secret number between 1 and 10!\n");
    printf("You have 3 guesses.\n");

    while (attempts < 3) {
        printf("Enter your guess: ");
        scanf("%d", &guess);

        attempts++;

        if (guess == secretNumber) {
            printf("Correct! You guessed the secret number!\n");
            break;
        }
        else if (guess > secretNumber) {
            printf("Too high!\n");
        }
        else {
            printf("Too low!\n");
        }
    }

    if (guess == secretNumber) {
        printf("You win!\n");
    }
    else {
```

```

    printf("You lose! The secret number was %d.\n", secretNumber);
}

return 0;
}

```

Claudes Code:

```
#include <stdio.h>
```

```

int main() {
    int secret = 7; // fixed secret number
    int guess;
    int attempts = 3;
    int won = 0;

    printf("Guess the secret number (between 1 and 10). You have %d tries.\n", attempts);

    for (int i = 1; i <= attempts; i++) {
        printf("Attempt %d/%d - Enter your guess: ", i, attempts);
        scanf("%d", &guess);

        if (guess == secret) {
            won = 1;
            break;
        } else if (guess > secret) {
            printf("Too high!\n");
        } else {
            printf("Too low!\n");
        }
    }

    if (won) {
        printf("Congratulations! You guessed the secret number.\n");
    } else {
        printf("Sorry, you're out of guesses. The secret number was %d.\n", secret);
    }

    return 0;
}

```

E). Code Analysis

Correctness:

Chatgpt- Sample answers work, displays correct screens, ends if guessed immediately. If a non-number is entered it immediately ends the game and uses all guesses and displays a you lose sign. It never displayed a screen telling me that I need to use letters and also used all guesses and ended the program so the error handling and edge cases are not good.

Claude- Sample answers work, displays correct screens, ends if guessed immediately. Once again if a non-number is entered the program immediately uses all guesses and displays a you lose sign and it had no display screen telling me to input numbers so the error handling and edge cases are bad.

Time Complexity:

Chatgpt- The time complexity is $3 * O(n) = O(3) = O(1)$. However, chatgpt uses a while loop which is typically slower than a for loop (claude).

Claude- The time complexity is $3 * O(n) = O(1)$. However, Claude uses a for loop which is typically faster than a while loop (chatgpt).

Space Complexity:

Chatgpt- The time complexity is the base time which is $O(1)$. However the use of 3 variables (chatgpt) is more space efficient then the use of 5 variables (claude).

Claude- The time complexity does narrow down to $O(1)$. However the use of 5 variables (claude) is less space efficient then the use of 3 variables (chatgpt).

Maintainability:

Chatgpt- Chatgpts variable names are very clear and used throughout, following the structure of the work is clear and simple. However, there are no comments either prologue or in line which could make the maintainability of the code over a long time hard. Code walkthrough also would have known that chatgpts code crashes when a non number is entered.

Claude- Claude has clear variable names and the structure is even more clear and easy to follow through and understand. However, there is only one in line comment and no prologue comments making the maintainability of this code difficult. Code walkthrough also would have known that claudes code crashes when a non number is entered.

F). Justification

I am picking Claudes code as my starting point because although it has a flaw in its 5 variables compared to 3 variables, it will be much easier to fix this problem then change the whole structure of the code. Also Chats code has a bug in its win/loss check if I changed its loop, the variable guess would become bugged and create more edge cases. The main flaw I need to fix in Claudes is by fixing the scanf to identify integers and repeat a guess if the input is a non integer which will get rid of the primary edge case and limit variable usage which is very easy.

G). Changes Made

- Removed the won flag as this saves space complexity and removes a variable
- Removed attempts variable as it saves space and removes another variable
- Adds a early exit and simplifies the overall logic
- Added a scanf validation to stop invalid inputs, clear them and reprompt with an error handling message to help the user.
- Added an improved loop counter that doesn't count bad inputs as attempts

Overall both codes had a simple version of the game. Chatgpts was much better for space complexity but lacked maintainability, time complexity and had problems with errors and edge cases. Claude however has a better structure, and maintainability. However it struggled with space complexity, errors and edge cases which was a much easier fix then Chatgpts. The new code now has correctness, and error handling, good space and time complexity even with the addition of the while loop as it is still a $O(1)$ and maintainability with the addition of multiple in line comments